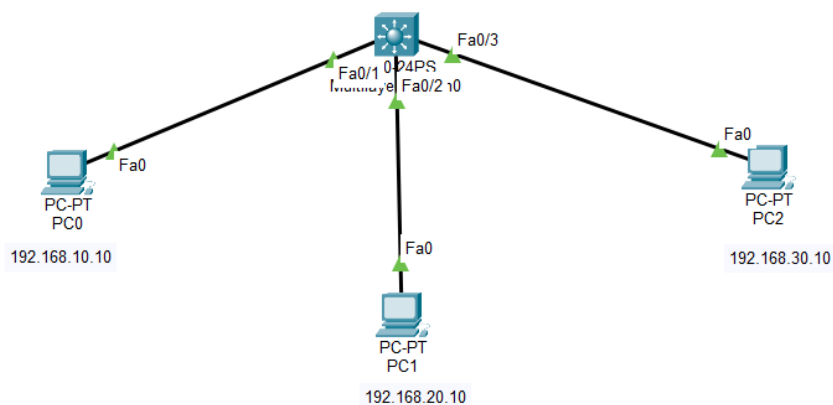


Layer 3 Switch Inter-VLAN Routing???? Theory: Using a Layer 3 Switch **No need for a router** **Uses SVIs (Switch Virtual Interfaces) for routing** **Faster than Router-on-a-Stick????** **Practical: Configuring Inter-VLAN Routing on a Layer 3 Switch????** **Hands-on Lab Enable IP Routing**

Task 1.

On a Cisco Layer 3 switch, create three VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support', VLAN 30 for 'Marketing') and assign at least one interface to each VLAN.



Switch#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	Sales	active	Fa0/1
20	Support	active	Fa0/2
30	Marketing	active	Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch#

Task 2.

Configure Switch Virtual Interfaces (SVIs) for each VLAN you created by assigning them unique IP addresses (e.g., 192.168.10.1/24 for VLAN 10, etc.) and ensure each SVI is in 'up/up' state.

Hint: Use the 'interface vlan' command to create SVIs and 'no shutdown' to enable them.

```
Vlan10          192.168.10.1    YES manual up          up
Vlan20          192.168.20.1    YES manual up          up
Vlan30          192.168.30.1    YES manual up          up
Switch#
Switch#
```

Task 3.

Enable IP routing on the Layer 3 switch so that devices in different VLANs can communicate with each other without using an external router.

Hint: Use the 'ip routing' global configuration command.

```
Switch(config)#ip routing
Switch(config)#do wr
Building configuration...
[OK]
Switch(config)#ex
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show running-config
Building configuration...

Current configuration : 1574 bytes
!
version 12.2(37)SE1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
no profinet
!
!
!
!
!
ip routing
!
!
!
!
!
.
```

Task 4.

Verify inter-VLAN routing by connecting two PCs (one in VLAN 10 and one in VLAN 20) and pinging between them. Capture and submit the ping results to show successful communication.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=2ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>
```

Task 5.

Refactor your VLAN and SVI configuration so that VLAN 30 uses a different subnet mask (e.g., /25 instead of /24). Explain what changes you made and how this affects communication with the other VLANs.

What does this change?

The /24 network had 254 usable host addresses.

The /25 network has 126 usable host addresses.

However, VLAN 30 can still communicate with VLAN 10 and VLAN 20 because each VLAN is a separate IP subnet and the Layer 3 switch routes between those subnets.

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- VLAN 30 (Marketing) was changed from a /24 subnet (255.255.255.0) to a /25 subnet (255.255.255.128).
- The SVI address remained 192.168.30.1, but its subnet mask was changed. The Marketing PC was also configured with the /25 mask. This reduces the number of usable host addresses from 254 to 126.
- Communication with VLAN 10 and VLAN 20 continues to work because the Layer 3 switch performs inter-VLAN routing between the different IP subnets.